



WE WANTED TO MOVE FROM HYDROGEN TO GREEN PROPULSION SYSTEM. AS A COMPANY, WAY BACK IN 2014-15, GETTING \$300-500 MILLION IN INDIA WAS AN INSANE THOUGHT ”

YASHAS KARANAM

Co-Founder and COO, Bellatrix

BELLATRIX

BELLATRIX'S JOURNEY BEGAN in 2011, when co-founders Rohan M. Ganapathy and Yashas Karanam were engineering students looking to build an innovative satellite technology that could earn them a patent and help them secure admission

to Ivy League universities.

Initially, the team experimented with rocket engines but soon realised the economics of reusable launch vehicles made the business impractical for a young Indian start-up. Building a competitor to SpaceX would have required hundreds of millions of dollars in investment.

“As a company, way back in 2014-15, getting \$300-500 million in India was an insane thought,” Karanam tells *BT* in an interview. Instead, they shifted focus to satellite propulsion, where they believed they could create differentiated technology.

The turning point came when Rohan's research paper on water-based satellite propulsion was selected for a NASA conference, where he met astronauts Neil Armstrong and Buzz Aldrin and visited NASA propulsion facilities. The project also caught the attention of former President A.P.J. Abdul Kalam, who encouraged the founders to build the technology in India and offered a recommendation letter.

“That letter somehow helped us raise money through industry CSR funds as use of water in electric propulsion of satellites was very novel,” says Karanam.

Their next milestone came when ISRO agreed to validate the technology and eventually awarded Bellatrix its first contract. The company was formally incorporated in 2015 to execute that order.

There was no turning back. After working with ISRO, the founders travelled internationally to meet satellite manufacturers and understand emerging market needs.

They observed that the satellite industry was rapidly moving from large geostationary satellites to smaller satellites operating in Low Earth Orbit (LEO) at altitudes of 600-1,000 kms.

Existing propulsion systems were designed for an earlier generation of satellites and were often bulky, expensive and heavy. Bellatrix decided to build a portfolio of compact propulsion systems tailored to this new generation of satellites.

One of Bellatrix's major goals has been replacing traditional toxic propellants with environmentally safer alternatives. “We wanted to move from hydrogen to green propulsion system,” says Karanam.

Bellatrix earns most of its revenue from exports, supplying propulsion systems directly to satellite manufacturers around the world. The Bengaluru-based start-up has raised approximately \$31 million in funding so far. **BT**

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